

Reconstruction of Two-Flavor Neutrino Oscillation Constraints from Early KamLAND Reactor Antineutrino Data

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Abstract

This paper discusses a two-flavor neutrino oscillation analysis of reconstructed reactor antineutrino data from the early KamLAND experiment (2002) using a (χ^2) based parameter estimation method. A combined rate and spectral shape likelihood was used to determine the oscillation parameters (Δm_{12}^2) and ($\sin^2 2\theta_{12}$). The best-fit values obtained are ($\Delta m_{12}^2 = 5.31 \times 10^{-5}$, eV²) and ($\sin^2 2\theta_{12} = 0.745$). Parametric contours at 68.3%, 90%, 95%, and 99% confidence levels are presented.

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1 Introduction

A significant discrepancy was historically observed between the predicted flux of neutrinos produced by nuclear fusion in the Sun and the number of neutrinos detected on Earth in experiments such as Homestake, SAGE, and GALLEX. This discrepancy, known as the “solar neutrino problem”[1], was eventually resolved through the discovery of neutrino oscillations. Neutrinos are produced in weak interactions as flavor eigenstates, which are quantum superpositions of mass eigenstates. Since the flavor eigenstates are not identical to the mass eigenstates, a neutrino produced in one flavor state can transform into another flavor state during propagation. This quantum mechanical phenomenon is known as “neutrino oscillation”[2]. For two-flavor oscillations in vacuum, the transition probability from a flavor state ν_α to ν_β is given by

$$P(\nu_\alpha \rightarrow \nu_\beta) = \sin^2 2\theta \sin^2(1.27\Delta m^2 \frac{L(\text{km})}{E(\text{GeV})}), \quad (1)$$

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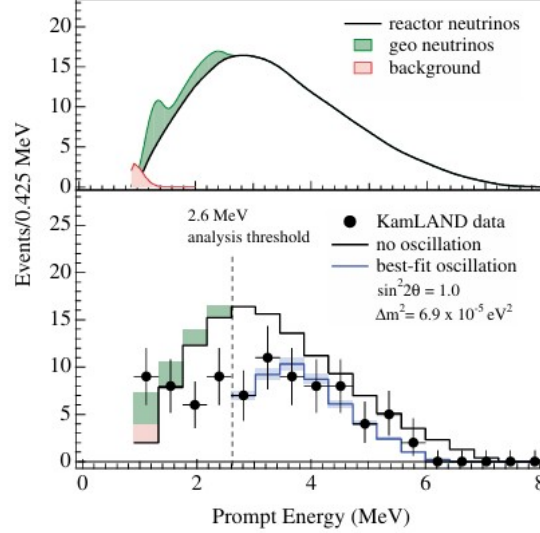


Figure 1: This figure has been obtained from [5]. Upper panel: Expected reactor $\bar{\nu}_e$ energy spectrum along with $\bar{\nu}_{geo}$ and background. Lower panel: Energy spectrum of the observed prompt events (solid circles with error bars), along with the expected no oscillation spectrum (upper histogram, with geo neutrino and background shown) and (lower histogram) best fit from [5] including neutrino oscillations. The shaded band indicates the systematic error in the best-fit spectrum. The vertical dashed line corresponds to the analysis threshold at 2.6 MeV.

where θ is the mixing angle, Δm^2 is the mass-squared difference in eV^2 , L is the baseline in km, and E is the neutrino energy in GeV [2]. The KamLAND experiment in Japan, which started in the year 2002, investigates neutrino oscillation by observing reactor electron antineutrinos, $\bar{\nu}_e$, emitted from distant nuclear power reactors. The 180 km long baseline enables KamLAND to address the solar neutrino problem. In the present work, we perform an independent reconstruction of two-flavor oscillation analysis of early KamLAND reactor antineutrino data in order to extract the best-fit values of Δm_{12}^2 and $\sin^2 2\theta_{12}$ through a χ^2 based parameter estimation framework.

2 The KamLAND Experiment

Kamioka Liquid Scintillator Anti-Neutrino Detector (KamLAND) [3] is located in the Kamioka Mine under Mount Ikonoyama, at former Kamiokande experiment[4] site at a depth of ~ 2700 m water equivalent. The data acquisition started in 2002, and has been operational over two decades now. About 79% of the neutrino flux in 2002 was obtained from 26 reactors within a distance ranging from 138 km to 214 km, making the baseline ~ 180 km. The construction and operational mechanism can be found in [5]. KamLAND is a anti-neutrino disappearance experiment and detects $\bar{\nu}_e$ in Liquid Scintillator (LS) through the inverse β decay reaction, $\bar{\nu}_e + p \rightarrow e^+ + n$ with a 1.8 MeV threshold[6]. The positron carries most of the energy of the anti neutrino and is annihilated quickly, providing a prompt event. Including annihilation, the detected energy for the positron is the sum of the kinetic energy and twice the rest energy. On average, the prompt energy can be written as, $E_{prompt} = E_{\bar{\nu}_e} - E_n - 0.8 \text{ MeV}$, where E_n is the average recoil energy of neutron. The first result, published in 2002, contains 162 ton.yr (145.1 d) of the reactor $\bar{\nu}_e$ data. The plot[5] for events per energy bin and prompt energy using this data is shown in figure 1.

Table 1: Digitized data for binned prompt energy distribution and event counts for KamLAND experiment and no oscillation case

Energy Bin (MeV)	E_{prompt} (MeV)	Events/bin	N_{MC}
2.600–3.025	2.81928	6.99231	16.4192
3.025–3.450	3.25301	11.0115	15.6154
3.450–3.875	3.66265	8.96538	13.6423
3.875–4.300	4.08434	7.94231	11.1577
4.300–4.725	4.53012	8.01539	9.33077
4.725–5.150	4.95181	3.92308	6.91923
5.150–5.575	5.38554	4.87308	5.01923
5.575–6.000	5.81928	1.95000	3.41154
6.000–6.425	6.21687	0.05000	2.16924
6.425–6.850	6.63855	0.12308	1.29231
6.850–7.275	7.07229	0.00000	0.48846

Table 2: Summary of systematic uncertainties contributing to the total estimated error[5]

Source of Uncertainty	Error (%)
Total LS mass	2.1
Fiducial mass ratio	4.1
Energy threshold	2.1
Efficiency of cuts	2.1
Live time	0.07
Reactor power	2.0
Fuel composition	1.0
Time lag	0.28
$\bar{\nu}$ spectra	2.5
Cross section	0.2
Total estimated error	6.4

3 Data Selection and Reconstruction

The selection cuts for the $\bar{\nu}_e$ can be found in [5, 7, 8]. Geologically produced antineutrinos from decay chains of U^{238} and Th^{232} in earth, called ‘geo neutrinos’ ($\bar{\nu}_{geo}$) produce events with prompt energy, $E_{\text{prompt}} < 2.49\text{MeV}$. This ambiguity can be avoided by requiring a prompt energy cut, $E_{\text{prompt}} > 2.6\text{MeV}$ [9] for the present analysis. The plot in figure 1 has been digitized using a web tool[10] and the reconstructed data is provided in table 1. The estimated systematic uncertainty (%) is shown in table 2. The ratio of observed reactor $\bar{\nu}_e$ events to expected in no oscillation case (R_{obs} with statistical error and systematic error) has been obtained from [5].

4 Chi Square Construction

The amount of matter between the detector and the reactors is small and thus, we don’t consider MSW effect in this analysis. Assuming CPT invariance, the survival probability of $\bar{\nu}_e$ for two flavor neutrino oscillation (assuming $\theta_{13} = 0$) in vacuum is,

$$P(\bar{\nu}_e \rightarrow \bar{\nu}_e) = 1 - \sin^2 2\theta_{12} \sin^2(1.27\Delta m_{12}^2 \frac{L(\text{km})}{E(\text{GeV})}) [2] \quad (2)$$

The ratio of the number of events detected in the KamLAND experiment and the expected number of events with no $\bar{\nu}_e$ disappearance is $(0.611) \pm (0.085)_{stat} \pm (0.041)_{syst}$ [5], where the second and third terms are error terms due to statistical and systematic errors. The ratio of the number of events from theoretical model and the expected number of events with no $\bar{\nu}_e$ disappearance is R_{th} . It is a function of $\sin^2 2\theta_{12}$ and Δm_{12}^2 . We construct,

$$\chi_{rate}^2 = \frac{0.611 - R_{th}(\sin^2 2\theta_{12}, \Delta m_{12}^2)^2}{0.085^2 + 0.041^2} \quad (3)$$

We construct the contribution from the prompt energy spectrum using a maximum likelihood method,

$\chi_{shape}^2 = -2 \ln(L_{shape})$, where L_{shape} is the likelihood function for the spectrum assuming Poisson statistical error. This can be evaluated using table 1 and statistical techniques from [11],

Let the fit parameters be

$$\boldsymbol{\eta} = (\sin^2 2\theta_{12}, \Delta m_{12}^2, \alpha).$$

For each energy bin E_i , the model probability is,

$$P_i = 1 - \sin^2 2\theta_{12} \left[\sin \left(1270 \Delta m_{12}^2 \frac{180}{E_i} \right) \right]^2 \quad (4)$$

The theoretical rate is

$$R_{th} = \frac{1}{n} \sum_{i=1}^n P_i.$$

The predicted number of events in each bin is

$$\mu_i = N_{mc} P_i.$$

Including the pull parameter α , this becomes

$$\mu_i(\alpha) = (1 + \alpha) \mu_i.$$

The rate contribution to the chi-square is,

$$\chi_{rate}^2 = 2 \frac{(R_{obs} - R_{th})^2}{\sigma_{stat}^2 + \sigma_{sys}^2} \quad (5)$$

The shape contribution with the pull term is

$$\chi_{shape}^2 = 2 \sum_{i=1}^n \left[\mu_i(\alpha) - N_i^{exp} + N_i^{exp} \ln \left(\frac{N_i^{exp}}{\mu_i(\alpha)} \right) \right] + \left(\frac{\alpha}{\sigma_\alpha} \right)^2. \quad (6)$$

We evaluate the oscillation parameters with the following χ^2 definition,

$$\chi^2 = \chi_{rate}^2 + \chi_{shape}^2 \quad (7)$$

This final χ^2 has been minimized to find the best-fit using [12] in python3. It is also provided as an ancillary file here.

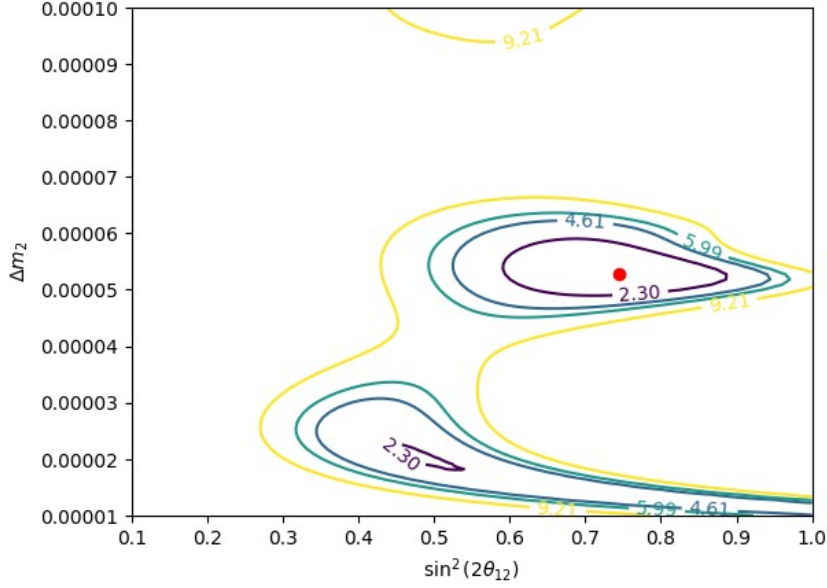


Figure 2: parametric contour plot for oscillation parameters $\sin^2 2\theta_{12}$ and Δm_{12}^2 with confidence levels for 68.3%, 90%, 95%, and 99%. The red point corresponds to the best-fit values.

5 Result

The KamLAND reactor antineutrino data were analyzed within the framework of two-flavor neutrino oscillations in vacuum, using a χ^2 based parameter estimation over the oscillation parameter space ($\sin^2 2\theta_{12}, \Delta m_{12}^2$). The best-fit point was obtained by minimizing the global χ^2 function across the scanned grid.

The minimum χ^2 value obtained is 5.707110284051756 (for 7 dof). The best fit Δm_{12}^2 is $5.30894583678182 \times 10^{-5} \text{ eV}^2$ and the best fit $\sin^2 2\theta_{12}$ is 0.744548661607577. The parametric contour plot is shown in figure 2. The corresponding confidence regions were constructed using the standard $\Delta\chi^2$ thresholds for two parameters. The contour levels at $\Delta\chi^2 = 2.30, 4.61, 5.99, 9.21$ correspond to confidence levels of 68.3%, 90%, 95%, and 99%, respectively.

Around $\Delta m_{12}^2 = 2 \times 10^{-5} \text{ eV}^2$, a secondary island exists which indicates a local minima, arising from the oscillatory structure of the reactor antineutrino survival probability. Such multiple allowed regions are characteristic of neutrino oscillation analyses, where different parameter combinations can produce comparable spectral distortions.

The confidence contours are observed to be more tightly constrained in the vertical direction than in the horizontal direction, indicating that the data place stronger constraints on Δm_{12}^2 than on the mixing parameter $\sin^2 2\theta_{12}$. This behavior is consistent with the strong sensitivity of the spectral distortion pattern to the oscillation phase, which depends directly on Δm_{12}^2 . The obtained best-fit region is consistent with the Large Mixing Angle (LMA) solution[5] to the solar neutrino problem.

6 Discussion

The data only from the first published result (2002) has been analyzed here. Thus, the data is not sufficient for 3 flavor neutrino oscillation analysis. Parameters for 3 flavor neutrino oscillation have been discussed in [ref.] The obtained values of $\sin^2 2\theta_{12}$ and Δm_{12}^2 from this paper are 0.74 and $5.3 \times 10^{-5} \text{ eV}^2$. The analysis does not include MSW effects, background events which have been discussed in [5]. Thus, a significant difference exists between the obtained values of the

parameters in this paper and [5] ($\sin^2 2\theta_{12} = 0.83, \Delta m_{12}^2 = 6.9 \times 10^{-5} eV^2$)

Code Availability

The Python code used for the numerical analysis and figure generation is provided as an ancillary file.

Acknowledgments

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